TITLE: Diabetes Risk Analysis using Data Visualization

Introduction

The diabetes prediction dataset contains health-related information used to analyse factors associated with diabetes. It includes variables such as age, gender, BMI, blood glucose level, hypertension, heart disease, and smoking history. By analysing and visualizing this data, meaningful patterns and relationships can be identified to better understand the factors influencing diabetes.

Objective of the Analysis

The primary objectives of this analysis are:

- To identify key factors influencing diabetes
- To explore relationships between health indicators and diabetes
- To visualize patterns and trends in the dataset
- To provide insights for early detection and preventive healthcare

Dataset Description

Source of Dataset:
The dataset used in this analysis is the *Diabetes Prediction Dataset* obtained from the data science platform Kaggle.

Number of Records:
The dataset contains **100,000 records** with **9 variables** representing different health indicators of individuals

- **Age** – Age of the individual
- **BMI** – Body Mass Index indicating body weight relative to height
- **Blood Glucose Level** – Blood sugar concentration
- **HbA1c Level** – Average blood sugar level over time
- **Hypertension** – Indicates presence of high blood pressure
- **Heart Disease** – Indicates presence of heart disease
- **Smoking History** – Smoking habits of individuals

- **Diabetes** – Indicates whether the individual has diabetes or not

These variables help in understanding patterns and potential risk factors associated with diabetes.

Exploratory Data Analysis

Initial exploration of the dataset was performed to understand:

- Structure and shape of the dataset
- Data types of variables
- Distribution of values
- Presence of missing or inconsistent data

This step helped in preparing the dataset for further analysis.

Starting Entries

	gender	age	hypertension	heart_disease	smoking_history	bmi	HbA1c_level	blood_glucose_level	diabetes
0	Female	80.0	0	1	never	25.19	6.6	140	0
1	Female	54.0	0	0	No Info	27.32	6.6	80	0
2	Male	28.0	0	0	never	27.32	5.7	158	0
3	Female	36.0	0	0	current	23.45	5.0	155	0
4	Male	76.0	1	1	current	20.14	4.8	155	0

Fig 1:Starting entries

Final Entries

	gender	age	hypertension	heart_disease	smoking_history	bmi	HbA1c_level	blood_glucose_level	diabetes
99995	Female	80.0	0	0	No Info	27.32	6.2	90	0
99996	Female	2.0	0	0	No Info	17.37	6.5	100	0
99997	Male	66.0	0	0	former	27.83	5.7	155	0
99998	Female	24.0	0	0	never	35.42	4.0	100	0
99999	Female	57.0	0	0	current	22.43	6.6	90	0

Fig 2:Final entries

Data Preprocessing and Cleaning

- Data preprocessing is an important step to ensure the accuracy and reliability of the analysis. The dataset was examined and prepared for visualization as follows:
- The shape of the dataset (number of rows and columns) was checked.
- Column names and data types were examined using df.columns and df.dtypes

- Binary columns like **diabetes ,heart disease and hypertension** were converted from 0/1 to meaningful labels (No/Yes) for better readability and visualization.
- Summary statistics were obtained using describe() for numerical columns.
- Missing values were identified using df.isnull().
- Missing values in numerical columns were replaced with the **mean of the respective column**.
- Ensured that numerical columns such as age, BMI, HbA1c, and blood glucose levels were in correct numeric format (int64, float64).

	gender	age	hypertension	heart_disease	smoking_history	bmi	HbA1c_level	blood_glucose_level	diabetes
0	Female	80.0	No	Yes	never	25.19	6.6	140	No
1	Female	54.0	No	No	No Info	27.32	6.6	80	No
2	Male	28.0	No	No	never	27.32	5.7	158	No
3	Female	36.0	No	No	current	23.45	5.0	155	No
4	Male	76.0	Yes	Yes	current	20.14	4.8	155	No

Fig 3:After replacing binary columns like diabetes ,hypertension and heart disease with meaningful label

	age	bmi	HbA1c_level	blood_glucose_level
count	100000.000000	100000.000000	100000.000000	100000.000000
mean	41.885856	27.320767	5.527507	138.058060
std	22.516840	6.636783	1.070672	40.708136
min	0.080000	10.010000	3.500000	80.000000
25%	24.000000	23.630000	4.800000	100.000000
50%	43.000000	27.320000	5.800000	140.000000
75%	60.000000	29.580000	6.200000	159.000000
max	80.000000	95.690000	9.000000	300.000000

Fig 4:Description of numeric column

Data Visualization

1.Gender Distribution in the Diabetes Dataset

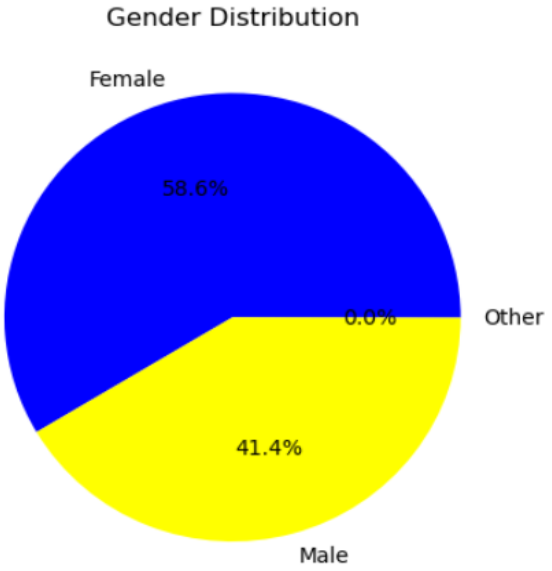


Fig 5: Gender distribution in dataset

- This pie chart represents the proportion of males and females.
- The dataset contains slightly more females than males.
- It helps in understanding gender balance in the dataset.

2. Age Distribution in the Diabetes Dataset

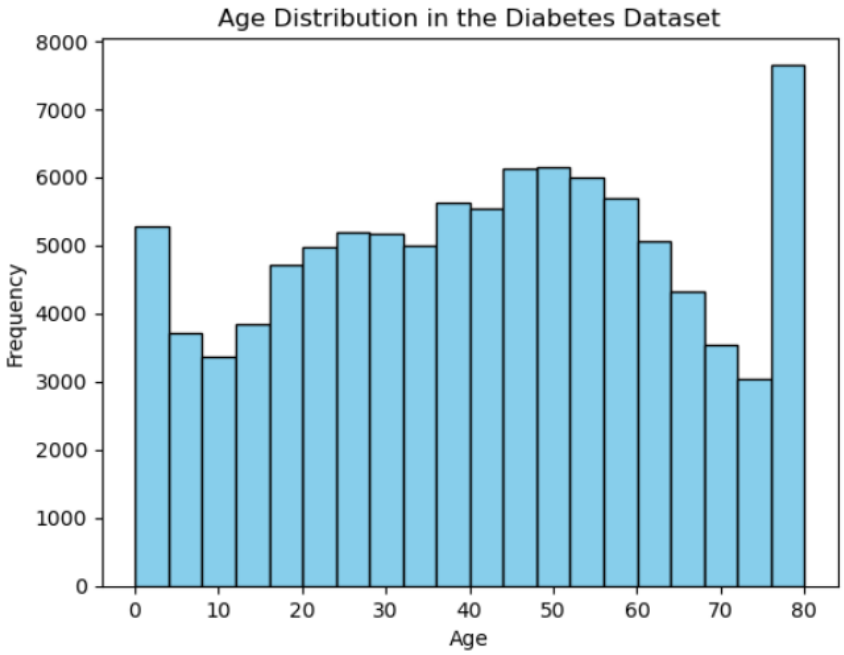


Fig 6:Age distribution in dataset

- This histogram shows how age is distributed among individuals.
- Most people fall in the middle-age group (around 30–60 years).

3. Total people suffering from diabetes:

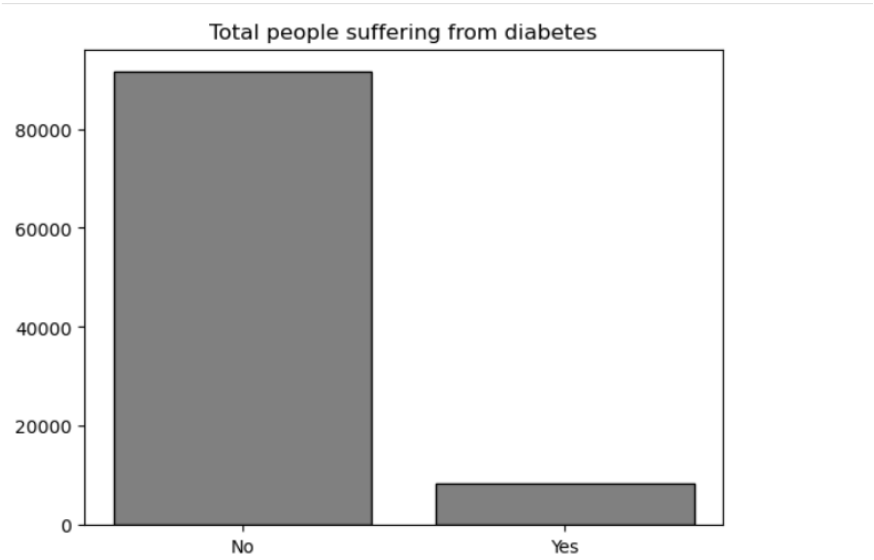


Fig 7:Total people suffering from diabetes

- This graph shows the number of people with and without diabetes.
- Majority of individuals do not have diabetes, indicating lower prevalence in the dataset.

4. Gender Distribution by Diabetes Status:

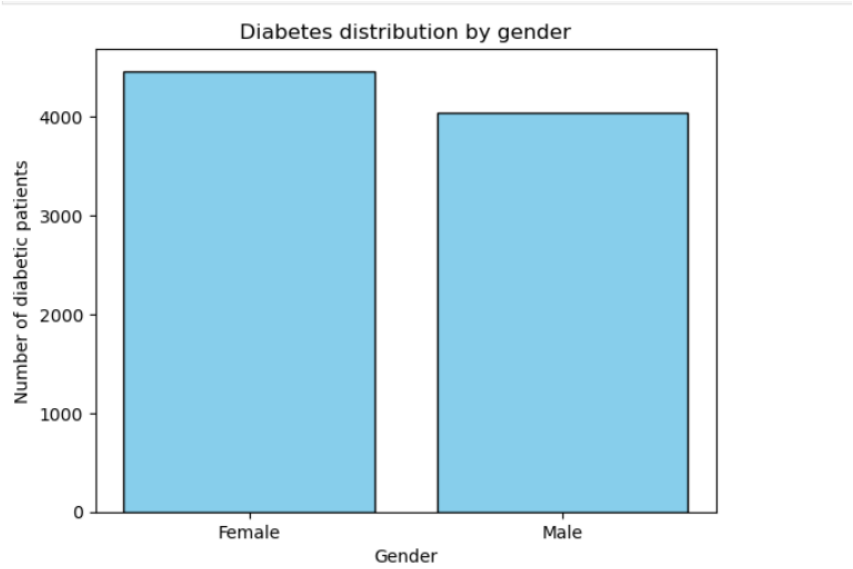


Fig 8:Gender distribution by Diabetes Status

- This graph compares diabetic patients across genders.
- Both males and females are affected, with slight variation in counts.
- It shows that diabetes is not limited to a single gender.

5. Heart Disease vs Diabetes

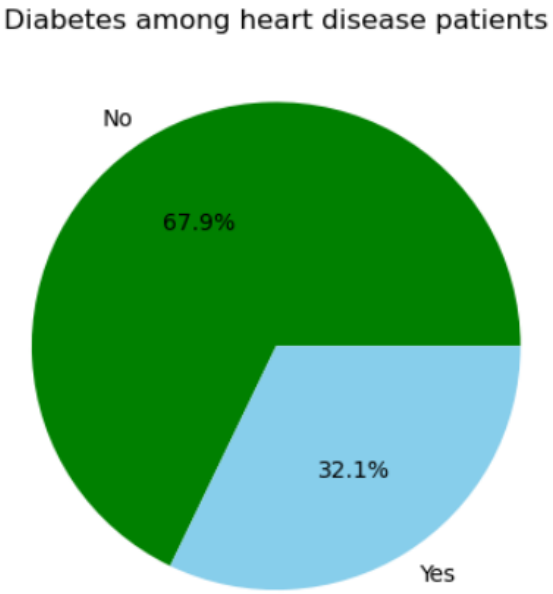


Fig 9:Diabetes among heart disease patients

- This chart shows how many heart disease patients also have diabetes.
- A significant portion of heart disease patients are diabetic.
- It suggests a possible relationship between heart disease and diabetes.

6. Smoking history vs frequency:

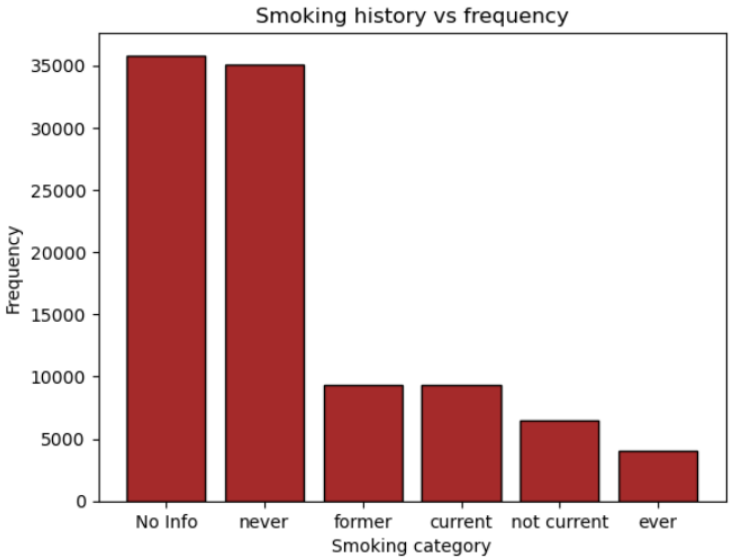


Fig 10:Smoking history vs frequency

- This graph displays the count of people in different smoking categories.
- Most individuals fall under “No info” and “never smoked”.
- It helps analyse lifestyle patterns related to health.

7. Diabetes among hypertension patients:

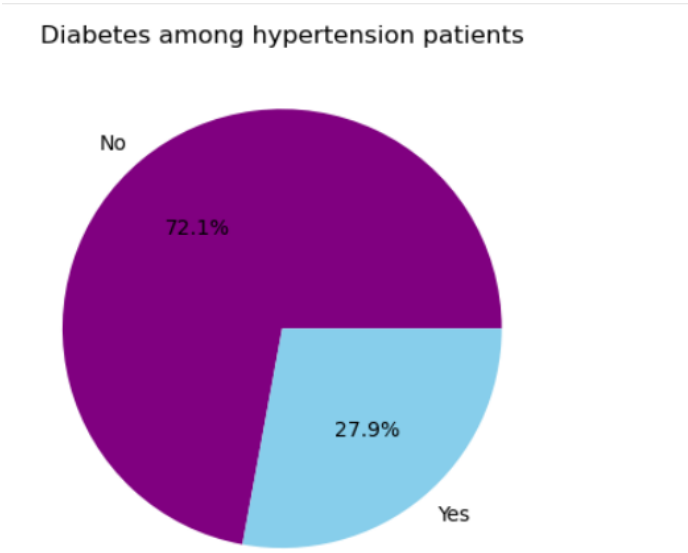


Fig 11:Diabetes among hypertension patients

- This chart shows diabetes presence among people with hypertension.
- A noticeable percentage of hypertensive patients have diabetes.
- It indicates a possible link between hypertension and diabetes.

8. BMI comparison by diabetes:

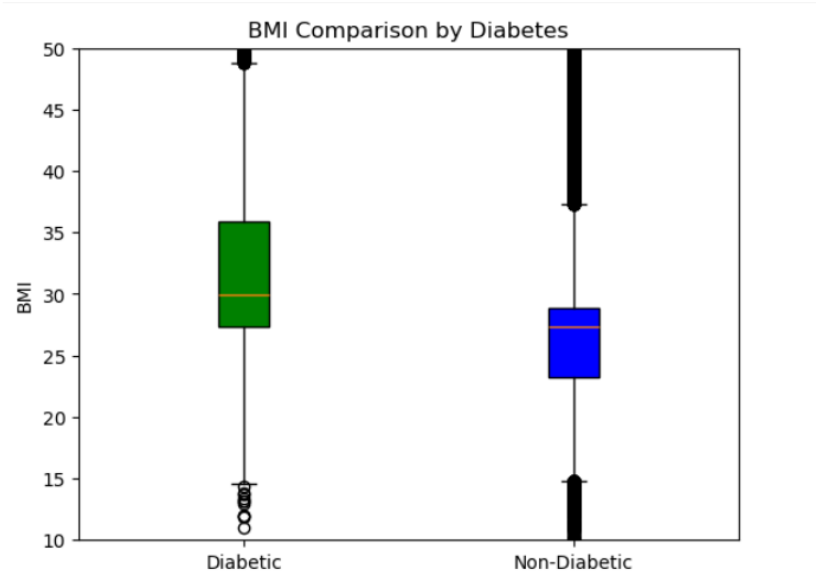


Fig 12:BMI comparison by diabetes

- This box plot compares BMI of diabetic and non-diabetic individuals.
- Diabetic patients tend to have slightly higher BMI values.
- It also highlights variation and outliers in BMI.

9. Age vs BMI:

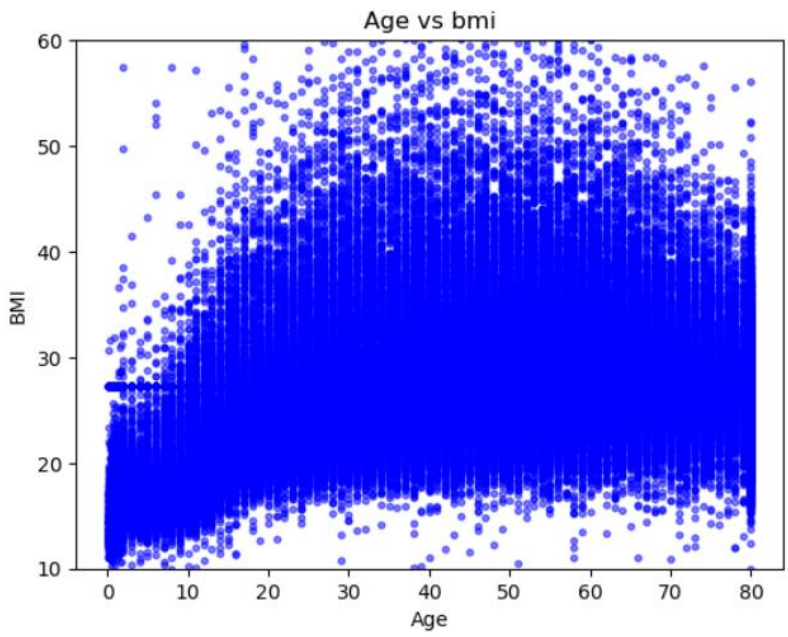


Fig 13:Age vs BMI

- This plot shows the relationship between age and BMI.
- There is a weak positive trend, but data is widely scattered.
- It indicates BMI is not strongly dependent on age alone.

10. Blood glucose vs diabetes

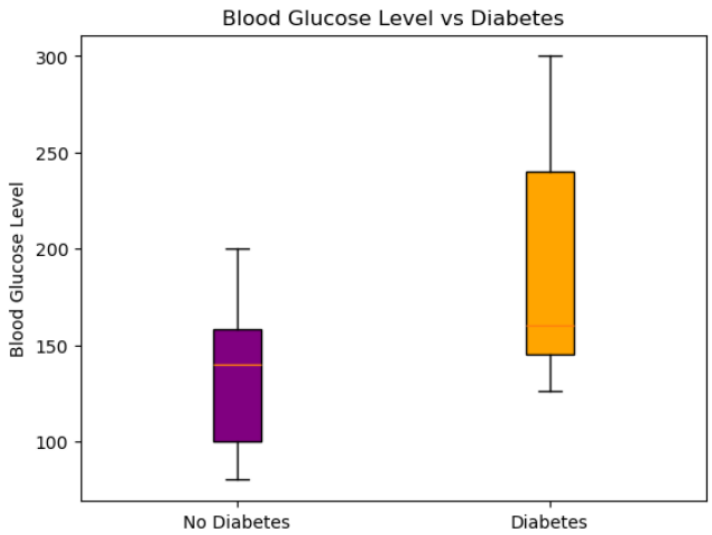


Fig 14:Blood glucose vs diabetes

- This box plot compares glucose levels for diabetic and non-diabetic individuals.
- Diabetic individuals have significantly higher glucose levels.
- It clearly shows glucose as a key indicator of diabetes.

11. Blood glucose distribution:

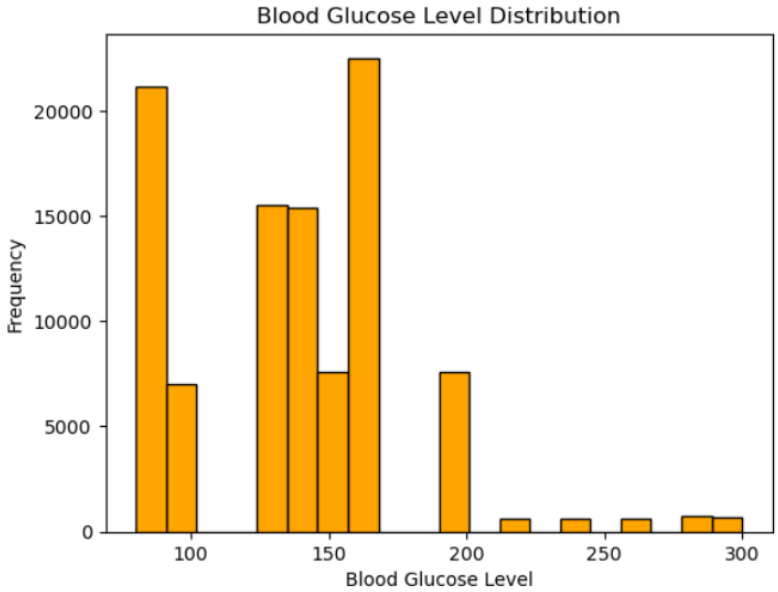


Fig 15:Blood glucose level distribution

- This histogram shows distribution of blood glucose levels.
- Most values lie in a moderate range, with some high-value outliers.
- It helps identify abnormal glucose levels in the dataset.

12. Correlation Heatmap of Health Variables

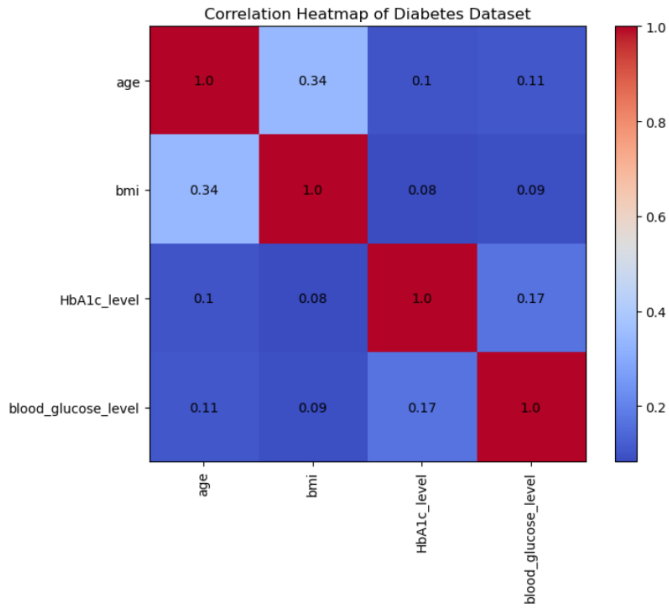


Fig 16:Correlation Heatmap of Health Variables

- This heatmap shows relationships between all numerical variables.
- Blood glucose and HbA1c have strong positive correlation but weak correlation with each other.
- It helps identify important features for prediction

Key Insights

1.Trends Observed:

- The dataset shows a low prevalence of diabetes, with the majority of individuals being non-diabetic.
- The age distribution is concentrated in the middle-age group (30–60 years), indicating that most individuals belong to the adult population.
- The dataset is skewed towards female participants, showing a higher proportion of females compared to males.

2.Relationships Between Variables:

- There is a strong relationship between blood glucose level and diabetes, making it a key indicator for prediction.
- Individuals with heart disease have a higher likelihood of having diabetes, indicating a positive association between the two conditions.
- Higher BMI values are observed in diabetic individuals, suggesting that obesity is linked to diabetes.

3.Interesting Patterns Discovered:

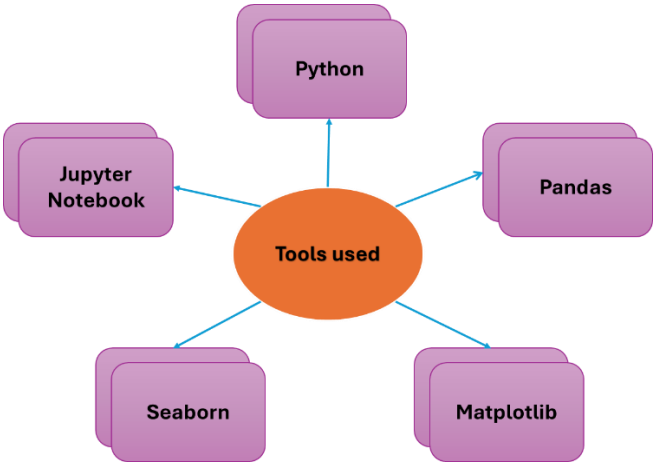
- Hypertension is commonly found alongside diabetes, suggesting that it may act as an associated risk factor.
- Diabetes occurrence is slightly higher among females, indicating possible gender-based variation in the dataset.
- A significant portion of smoking history data is labelled as “No Info,” which may affect the reliability of analysis related to smoking.

Tools Used

Explanation:

- **Python**: Used as the main programming language for data analysis and visualization
- **Jupyter Notebook:** Provided an interactive environment to write, run, and document code.
- **Seaborn:** Used for advanced and visually appealing statistical visualizations.
- **Matplotlib:** Used to create basic graphs and visualizations.

- **Pandas:** Used for data cleaning, preprocessing, and handling structured datasets.



Conclusion:

- Diabetes occurrence in the dataset is influenced by key factors such as blood glucose level, HbA1c, BMI, and hypertension, with blood glucose and HbA1c being the strongest indicators.
- Individuals with heart disease and higher BMI show a greater likelihood of having diabetes, indicating the presence of important health risk associations.
- The dataset highlights the need for better data quality in certain variables (such as smoking history) to improve the accuracy and reliability of analysis.

Reference

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